PrioraFlow — Owner Manual

Version: 1.1
Date: May 2026
For: Service center owners, general managers, and operations leadership

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1. What Is PrioraFlow?

PrioraFlow is an **AI-powered flow-control system for the entire service center** — reception, workshop, parts, and delivery.

It is not a generic task manager. It is not a simple job tracker. PrioraFlow answers the critical question every service center faces daily:

> ***Which vehicle needs attention right now — and what should be done about it?***

The system gives **management full visibility** over every job, every stage, and every delay. It turns reactive workshop operations into a proactive, priority-driven flow.

The Problem It Solves

In a busy service center, managers oversee 15–50+ vehicles across reception, workshop, and delivery. Without a smart system:

- Promised delivery times get missed
- Customers sit waiting without updates
- Parts delays go unnoticed
- High-value or VIP customers don't get prioritized
- Vehicles sit idle because nobody noticed they were stuck
- Workshop capacity is underutilized
- Management has no real-time visibility into what's happening

PrioraFlow solves this by giving every job a **live priority score**, explaining **why** it's urgent, telling the team **exactly what to do next**, and giving **management a clear dashboard** of the entire operation.

Management Value

PrioraFlow gives owners and managers what they've never had before:

- **Real-time visibility** — See every job, every delay, every risk in one place. No more walking the floor.
- **Priority-driven operations** — The system tells the team what to focus on, based on data, not gut feeling.
- **Revenue capture** — Deferred work tracking means no lost revenue. Customer approval flow means no ambiguity.

- **Accountability** — Every status change is logged. Every action has an owner. Every delay is visible.
- **Scalability** — Works with 5 jobs or 500. The priority engine and next actions scale with the operation.

Key Differentiators

Feature	Traditional Workshop Software	PrioraFlow
Priority scoring	None — all jobs equal	Live 0–100 score, auto-updated
What to do next	Each person figures it out	Next Best Action for every job
Customer follow-up	Manual tracking	Auto-flagged when waiting or overdue
Promised date risk	Manual mental note	Auto-calculated, flagged before it's late
Workshop ↔ Overall sync	Manual dual entry	Auto-synced, impossible to break
Parts delay visibility	Separate system	Integrated into priority score
Customer Informed flow	Sticky notes, WhatsApp groups	One-click with priority adjustment
Management visibility	Walk the floor or ask	Real-time board with priority, risk, and actions

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2. Who Is It For?

Primary Decision Makers

- **Service Center Owners** — Full visibility into every job, every delay, every missed promise. ROI-driven oversight.
- **General Managers / Operations Managers** — Real-time dashboard of the entire service center: reception, workshop, and delivery.
- **Workshop Controllers** — See the full workshop flow, identify bottlenecks, allocate resources.

Active Users

- **Service Advisors** — Manage their jobs, priorities, next actions, and customer follow-ups.
- **Technicians** — Receive assignments, complete inspections, report progress.
- **Parts Coordinators** — Track parts status, update ETAs, unblock workshop.
- **Administrators** — Configure the system, manage users, and customize priority weights.

Ideal Service Center Profiles

- Premium and luxury automotive brands (Mercedes-Benz, BMW, Audi, etc.)
- Service centers with 10+ active repair orders per advisor
- Operations where missed promised dates cost real money and customer retention
- Teams that want data-driven decisions, not guesswork
- Management that needs real-time visibility, not just end-of-day reports

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3. Getting Started

First Login

1. Open PrioraFlow in your browser (e.g., `https://your-workshop.prioraflow.com`)
2. Log in with your admin credentials (provided during setup)
3. You'll land on the **Job Board** — your main cockpit

Initial Setup

As an admin, configure these before your team starts using the system:

1. **Users & Roles** — Add your advisors, technicians, and managers (Settings → Users)
2. **Labour Rates** — Set your hourly rates for different operations (Settings → Labour Rates)
3. **Priority Weights** — Customize scoring to match your workshop's priorities (Settings → Priority Matrix)
4. **Inspection Templates** — Create or customize inspection checklists (Admin → Templates)
5. **Integrations** — Connect webhooks for notifications (Settings → Integrations)

Quick Tour

Screen	What It Shows
Job Board	All active jobs across the entire service center, sorted by priority
Job Detail	Full information for one job: status, priority, estimate, inspection, actions
Priority Snapshot	Top urgent jobs + next actions sidebar
Settings	System configuration, priority weights, users, labour rates
Admin	Dashboard stats, booking import, template management

Management sees everything. Advisors see their own jobs. Technicians see their assignments. This role-based visibility gives owners the full picture while keeping each team member focused on their work.

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4. User Roles & Permissions

Role	Can See	Can Do
Admin	Everything	Full system access: settings, users, templates, all jobs
Manager	All jobs	View and manage all jobs, configure settings
Service Advisor	Their assigned jobs	Create jobs, update status, send estimates, inform customers
Technician	Jobs assigned to them	Complete inspections, update workshop stage, upload media

Role-Based Job Visibility

- Advisors see **only their own jobs** by default
- Technicians see **only jobs they're assigned to**
- Managers and admins see **all jobs**

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5. The Job Lifecycle

Every vehicle that enters the workshop follows a structured lifecycle. The system enforces valid transitions — you cannot skip steps or jump backwards arbitrarily.

Status Flow Diagram

```
┌──────────┐
│  BOOKED  │  ← Vehicle scheduled, not yet received
└────┬─────┘
     │
     ▼
┌──────────┐
│ CHECKING │  ← Vehicle received, inspection/diagnosis in progress
└────┬─────┘
     │
     ▼
┌──────────────┐
│ ESTIMATE SENT│  ← Quote sent to customer, awaiting decision
└────┬─────┬───┘
     │     │
     │     └──────────┐
     ▼                │
┌──────────┐          │
│ APPROVED │ ←────────┘   Customer approved the work
└────┬─────┘
     │
     ▼
┌─────────────┐
│ IN PROGRESS│  ← Workshop actively working on the vehicle
└────┬───┬────┘
     │   │
     │   └──────────────┐
     ▼                  │
┌───────────────┐       │
│ WAITING PARTS │ ←─────┘   Parts needed, work paused
└────┬──────────┘
     │
     ▼
┌──────────────┐
│ QUALITY CHECK│  ← Work complete, final QC in progress
└────┬────┬────┘
     │    │
     │    └──────┐
     ▼           │
┌─────────┐      │
│  READY  │ ←────┘   Vehicle ready for delivery
└────┬────┘
     │
```

```
      ▼
 ┌────────┐
 │ CLOSED │  ← Vehicle delivered, job archived
 └────────┘
```

Allowed Transitions

From	Can Move To
Booked	Checking, Closed
Checking	Estimate Sent, Approved, In Progress, Closed
Estimate Sent	Checking, Approved, Closed
Approved	Estimate Sent, In Progress, Closed
In Progress	Waiting Parts, Quality Check, Ready, Closed
Waiting Parts	In Progress, Closed
Quality Check	In Progress, Ready
Ready	Quality Check, Closed
Closed	*(terminal — no transitions out)*

Three Operational Phases

Phase	Statuses	Who Owns It
Reception / Advisor	Booked, Checking, Estimate Sent, Approved	Service Advisor
Workshop	In Progress, Waiting Parts, Quality Check	Workshop / Technician
Delivery	Ready, Closed	Service Advisor

Side Effects on Status Changes

The system automatically handles these when status changes:

Action	What Happens
Move to **Ready**	`completed_at` timestamp is set
Move to **Closed**	`invoiced_at` timestamp is set

Action	What Happens
Move **away from Closed**	`invoiced_at` and `archived_at` are cleared
Move **away from Ready** (backwards)	`completed_at` is cleared
Move to **Waiting Parts**	Workshop Stage is cleared; Parts Status auto-sets to "Order Parts"
Parts Status → **Parts Ready**	Workshop Stage auto-sets to "Waiting to Start"
Workshop Stage → **WIP**	Overall status auto-sets to "In Progress"
Workshop Stage → **Quality Check**	Overall status auto-sets to "Quality Check"
Workshop Stage → **Ready Handover**	Overall status auto-sets to "Ready"

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6. The Priority Engine

This is the heart of PrioraFlow. Every active job receives a **live priority score** from 0 to 100.

Priority Levels

Score	Level	Color	Meaning
0–39	**Low**	Grey	No urgency. Normal progress.
40–64	**Normal**	Blue	Standard attention needed.
65–84	**High**	Amber	Elevated urgency — action soon.
85–100	**Critical**	Red	Immediate action required.

How the Score Is Calculated

Formula: `Priority = min(100, 10 + Σ(active factor weights))`

Every job starts with a base score of 10. Then each active factor adds its weight. The total is capped at 100.

Priority Factors

Factor	Default Weight	When It Applies	Category
Promise Overdue	30	Promised date has passed and job is not Ready/Closed	Promise Risk
Promise Due ≤ 2h	20	Promised date is within 2 hours	Promise Risk
Promise Due ≤ 6h	10	Promised date is within 6 hours	Promise Risk
No Promised Date	5	Active job (not booked/closed) has no promised date	Promise Risk
Customer Waiting	22	Customer is marked as waiting at the workshop	Customer
Customer Angry	18	Customer sensitivity set to "Angry"	Customer
Customer VIP	16	Customer sensitivity set to "VIP"	Customer
Customer Comeback	14	Customer sensitivity set to "Comeback"	Customer
Waiting Customer Decision	20	Job is in "Estimate Sent" status	Approval
Parts Backorder	22	Parts status is "Backorder"	Parts Risk
Parts Waiting Warehouse	16	Parts status is "Waiting Warehouse"	Parts Risk
Parts Need Order	12	Parts status is "Order Parts" or job is "Waiting Parts"	Parts Risk
Idle 12h+	12	Job hasn't been updated in 12+ hours (not booked)	Idle Risk
Idle 6h+	6	Job hasn't been updated in 6+ hours (not booked)	Idle Risk
Stage: Checking/Diagnosis	10	Job is in "Checking" status	Stage Urgency
Stage: QC / Near Delivery	10	Job is in "Quality Check" status	Stage Urgency
Ready to Inform	20	Job is "Ready" but customer not yet informed	Delivery
High Estimate Value	8	Estimate total ≥ 10,000	Value

Factor	Default Weight	When It Applies	Category
Medium Estimate Value	4	Estimate total ≥ 5,000	Value

Weight Scale Guide

Range	Intent	Examples
4–8	Mild nudge — nice to address	Medium estimate, idle 6h
10–16	Moderate — should act soon	Parts waiting, VIP customer, QC stage
18–22	Strong — act now	Customer waiting, backorder, ready to inform
30	Maximum — critical alert	Promise overdue

Example Priority Calculation

Job: Mercedes C-Class, RO-49126

Factor	Weight
Base	10
Promise Due ≤ 2h	20
Customer Waiting	22
Customer VIP	16
Total	**68 → High**

Priority explanation shown to advisor:

> *Promise risk: due ≤2h +20 · Customer waiting +22 · Customer sensitivity: VIP +16*

What Gets Zeroed by "Customer Informed"

When a job is Ready and you click "Customer Informed," urgency factors about getting the car ready or informing the customer are zeroed:

Zeroed (urgency factors):

- Promise risk (overdue, due soon)
- Customer waiting
- Waiting customer decision
- Parts risk
- Idle risk
- Stage urgency (checking, QC)
- Ready to inform

Kept (permanent facts):

- Customer sensitivity (angry, VIP, comeback)
- Estimate value (high, medium)

This means an informed VIP customer with a high-value estimate still shows a meaningful priority — the car is handled, but the customer relationship matters.

Booked Jobs and Idle Risk

Booked appointments are excluded from idle risk scoring. A booked car that hasn't moved in 12 hours is normal — it's an appointment waiting for its slot, not a stuck job.

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7. Next Best Actions

Every active job has a **Next Best Action** — the system tells the responsible person exactly what to do. This eliminates the "walk the floor and ask" approach and gives management confidence that nothing falls through the cracks.

How It Works

The action is determined by the job's current status and phase. Risk signals (promise overdue, customer waiting, VIP, etc.) increase the action's urgency and explain *why* it matters now. **Management can see the full action queue across all advisors**, not just per-person.

Actions by Status

Job Status	Next Best Action	Owner
Booked	Receive vehicle and start check-in	Advisor
Checking	Complete diagnosis and prepare estimate	Advisor

Job Status	Next Best Action	Owner
Estimate Sent	Follow up customer decision	Advisor
Approved	Print job card and release to workshop	Advisor
Waiting Parts	Check parts ETA and update plan	Parts
In Progress (no technician)	Assign technician and start work	Workshop
In Progress (approval blocker)	Resolve advisor/approval blocker	Advisor
In Progress (parts blocking)	Check parts ETA and unblock technician	Parts
In Progress (normal)	Check technician progress	Workshop
Quality Check	Complete QC and prepare delivery	Workshop
Ready	Notify customer for collection	Advisor

Action Urgency Levels

Urgency	When	Color
Low	Base scenario, no risk signals	Grey
Normal	Some risk signals present	Blue
High	Multiple risk signals	Amber
Critical	Promise overdue, angry customer, or multiple strong signals	Red

Risk Signals

Each action shows the risk signals that make it urgent:

- **Promise overdue** — promised date has passed
- **Promise due within 2h** — delivery promised very soon
- **Promise due within 6h** — delivery promised today
- **Customer waiting** — customer is at the workshop
- **Angry/VIP/Comeback customer** — sensitive customer relationship
- **Idle 6h+/12h+** — job hasn't progressed in too long
- **Parts: backorder/waiting/order** — parts are blocking progress
- **High/medium estimate value** — significant financial value at stake

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8. Workshop Stages

Workshop stages track the **physical progress** of the vehicle inside the workshop. They are separate from the overall job status.

Workshop Stage Flow

```
Waiting to Start → Diagnosis → Estimate Prep → Advisor/Approval → WIP → Final
```

Workshop Stages Detail

Stage	Label	What It Means
`waiting_technician`	Waiting to Start	Vehicle received, waiting for a technician or bay
`diagnosis`	Diagnosis	Technician is inspecting and diagnosing the vehicle
`estimate_prep`	Estimate Prep	Technician or advisor is preparing the quote
`customer_approval`	Advisor / Approval	Waiting for advisor follow-up or customer decision
`work_in_progress`	WIP	Technician is actively performing the repair
`final_test`	Final Test	Road test or final functional test
`quality_check`	QC	Quality control check in progress
`ready_handover`	Ready Handover	Vehicle is ready for customer delivery

Workshop ↔ Overall Auto-Sync

When you change the Workshop Stage, the overall job status updates automatically:

Workshop Stage Change	Overall Status Auto-Set To
→ WIP	In Progress
→ Quality Check	Quality Check
→ Ready Handover	Ready

And when overall status changes:

Overall Status Change	Parts Status Auto-Set To
→ Waiting Parts	Order Parts

Parts Status Change	Workshop Stage Auto-Set To
→ Parts Ready	Waiting to Start

When Workshop Stage Is Disabled

The Workshop Stage dropdown is disabled (greyed out) when the vehicle is not in the workshop:

Overall Status	Workshop Stage	Parts Status
Booked, Checking, Estimate Sent, Approved	❌ Disabled	❌ Disabled
Waiting Parts	❌ Disabled	✅ Active
In Progress, Quality Check, Ready	✅ Active	✅ Active

When disabled, the Operational Snapshot shows "—" instead of a misleading stage name.

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9. Parts Status Flow

Parts status tracks the supply chain state for parts needed by the job.

Parts Statuses

Status	Label	What It Means
`no_parts`	No Parts	No parts needed for this job
`order_parts`	Order Parts	Parts need to be ordered
`waiting_warehouse`	Waiting Warehouse	Parts ordered, waiting at warehouse
`backorder`	Backorder	Parts are on backorder — delayed supply
`parts_ready`	Parts Ready	All parts received and available

Auto-Sync Behaviors

- Overall status → **Waiting Parts**: Parts status auto-sets to **Order Parts**
- Parts status → **Parts Ready**: Workshop stage auto-sets to **Waiting to Start**

Parts Status and Priority

Parts status directly affects the priority score:

Parts Status	Priority Impact
Backorder	+22 (strong — supply chain is stuck)
Waiting Warehouse	+16 (moderate — parts on the way)
Order Parts	+12 (mild — needs action but not blocked yet)
No Parts / Parts Ready	+0

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10. Customer Informed System

When a vehicle is ready for delivery, the most important action is informing the customer. The Customer Informed system makes this a one-click operation with smart priority adjustment.

When the Button Appears

The **"Customer Informed"** button appears:

- On the job detail page (in the status section)
- On the job board cards
- **Only when** the job status is **Ready**
- **Only when** `customer_informed` is `false`

What Happens When You Click It

1. `customer_informed` is set to `true`
2. `is_customer_waiting` is auto-cleared (customer was waiting, now they know)
3. Urgency factors are zeroed from the priority score
4. Button changes to ✓ **Informed** (disabled, cannot be unclicked)
5. Toast notification confirms: "Customer informed — urgency factors cleared from priority"

Priority Before and After — Example

Before clicking "Customer Informed":

Factor	Weight
Base	10
Ready to Inform	20
Customer Waiting	22
Promise Due ≤ 2h	20
Total	**72 → High**

After clicking "Customer Informed":

Factor	Weight
Base	10
(urgency factors zeroed)	0
Customer VIP (permanent)	16
High Estimate Value (permanent)	8
Total	**34 → Low**

The priority drops because the car is handled and the customer knows. But permanent facts (VIP, value) keep a small baseline — you still treat VIP customers well.

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11. The Job Board

The Job Board is the main screen — a Kanban-style view of all active jobs across the entire service center, organized by status. **This is management's real-time visibility tool**: see the whole operation at a glance, identify bottlenecks, and know exactly where attention is needed.

Board Layout

The board is divided into **three phases**, each with its own color:

Phase	Color	Columns
Reception / Advisor	Blue	Booked, Checking, Estimate Sent, Approved
Workshop	Orange	In Progress, Waiting Parts, Quality Check
Delivery	Green	Ready, Closed

What Each Card Shows

- **Job number** and current status badge
- **Customer name** and **vehicle** (make, model, plate)
- **Priority score** (color-coded badge)
- **Priority reasons** (expandable on hover/click)
- **Parts status** badge (if applicable)
- **Customer sensitivity** badge (VIP, Angry, Comeback)
- **Promised date** with overdue/due-soon indicators
- **Customer Informed** button (when Ready and not yet informed)

Priority Sidebar

The sidebar shows:

1. **Priority Snapshot** — Top 6 most urgent jobs with scores and next actions
2. **Advisor Actions** — Top 8 recommended actions, sorted by priority + action urgency

Drag and Drop

Jobs can be moved between columns by changing status via the dropdown or the "Next →" button. The board updates in real time.

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12. Job Detail Page

Clicking a job opens its full detail page with everything about that repair order.

Sections

Operational Snapshot

At-a-glance summary cards:

Card	Shows
Current Status	Overall job status with color badge
Workshop Stage	Current workshop stage (or "—" if not applicable)
Parts Status	Current parts status
Priority Flags	Customer waiting/not waiting + sensitivity level
Advisor	Assigned service advisor
Technician	Assigned technician
Odometer	Vehicle odometer reading (synced from inspection)

Customer & Vehicle Info

- Customer name, phone, email
- Vehicle make, model, year, color, plate, VIN

Status Controls

- **Overall Status dropdown** — shows current status, auto-changes on selection
- **"Next →" button** — advances to the logical next status
- **Workshop Stage dropdown** — disabled when not applicable
- **Parts Status dropdown** — disabled in reception phases
- **Promised Date** picker — setting a promised date removes the +5 "No Promised Date" penalty
- **Customer Informed** button — appears when Ready

Priority Explanation

Shows the full breakdown:

```
Priority: 72 / High
• Promise risk: due ≤2h +20
• Customer waiting +22
• Customer sensitivity: VIP +16
• Ready to inform customer +20
```

Estimate / Quote

- Full estimate builder with labour, parts, and sublet lines
- Group by concern area or general/other

- Tax calculation (auto)
- Send approval link to customer
- Track customer decisions

Inspection

- Link to inspection workspace
- Odometer and fuel level readings
- Findings with severity markers
- Photos and media

Media Gallery

- All photos and documents attached to the job
- Upload new media
- View/delete existing media

Job History

- Timestamped log of all status changes
- Shows who changed what and when

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13. Inspection System

Overview

Every job can have one inspection. The inspection captures the vehicle's condition when it arrives and documents findings that may become estimate lines.

Inspection Templates

Templates define the structure of an inspection. Each template contains:

- **Sections** — Groups like "Exterior", "Interior", "Engine", "Underbody"
- **Items** — Individual checks within each section

Item Input Types

Type	What It Captures	Example
`pass_fail`	Green/Red pass or fail	Brake pads — Pass / Fail
`yes_no`	Yes or No	Tyre rotation needed?
`ok_warn_fail`	Three-level: OK / Warning / Fail	Belt condition
`number`	Numeric value	Tyre tread depth (mm)
`odometer`	Mileage reading (km)	Vehicle odometer
`fuel_level`	Fuel level	Fuel gauge reading
`text`	Free-text notes	Additional observations
`toggle`	On/off switch	AC working?
`photo`	Photo capture	Damage documentation

Odometer Sync

When a technician enters the odometer reading in the inspection, the value **automatically syncs to the job's odometer field**. No double entry needed.

Inspection Lifecycle

```
In Progress (editable) → Submitted (locked) → Reviewed → Approved
```

- **Creating** an inspection auto-moves a Booked job to Checking
- Once submitted, the inspection is locked and cannot be edited
- An advisor can **reopen** a submitted inspection to allow corrections
- Reopen moves the inspection back to In Progress

Urgency Markers

Each inspection response can be marked with urgency:

Urgency	Meaning
None	Normal finding
Low	Minor note
Medium	Should be addressed
High	Must be addressed — safety or customer impact

Urgency	Meaning
Critical	Immediate attention required

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14. Estimate Builder & Authorization

Estimate Builder

The estimate builder creates the quote for the customer. It supports:

- **Labour lines** — Operation code, hours, labour rate, line total
- **Parts lines** — Part number, quantity, unit price, line total
- **Sublet lines** — External work sent to third parties
- **Grouping** — Lines can be grouped by concern area (linked to inspection findings) or as General/Other

All financial totals are calculated on the **backend** — the frontend cannot override prices.

Customer Approval Flow

The customer approval flow is a key differentiator for management: no more verbal approvals, no more "I think they said yes." Every decision is tracked, timestamped, and auditable.

1. Advisor creates the estimate
2. Advisor clicks **"Send Approval"** — system generates a unique approval link
3. Link is sent to the customer (via WhatsApp, SMS, or email)
4. Customer opens the portal and sees:
   - Job and vehicle summary
   - Inspection findings (grouped by severity: red → amber → green)
   - Estimate lines with approve/decline/defer buttons
   - Job photos
5. Customer makes decisions per line
6. System records decisions and:
   - Approved lines → proceed with work
   - Declined/deferred lines → create **Deferred Work** records
   - If all lines approved → job auto-moves to **Approved**

7. Advisor receives notification of customer decision

Approval Links

- Links are hashed (SHA-256) for security — raw tokens are never stored
- Links expire after 7 days by default
- Links are single-use — once the customer submits, the link is invalid
- Links can be revoked manually by the advisor

Estimate Line History

Before any estimate line is modified, the system snapshots the previous version into `estimate_line_history`. This provides a full audit trail of quote changes.

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15. Booking Import

Overview

The booking import allows bulk creation of jobs from a DMS export file (XLSX or CSV).

How to Use

1. Go to **Admin → Booking Import**
2. Upload your XLSX or CSV file
3. Map columns to system fields
4. Preview and confirm import

Supported Fields

Column	Maps To
Customer name	Customer first/last name
Contact / Phone	Customer phone
Vehicle Make/Model/Year	Vehicle details
Plate number	Vehicle plate
VIN	Vehicle VIN
Appointment date	Job promised date

Smart Features

- **Header row detection** — Rows with "Customer name", "Name", "Contact" etc. are automatically skipped
- **Duplicate detection** — Existing customers/vehicles are matched and reused
- **Auto-booking** — Imported jobs start as "Booked" status
- **Odometer import** — Mileage values from the booking file are stored

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16. Media & Photo Management

Upload Methods

- **Direct upload** — File uploaded through the browser
- **Presigned URL** — For large files or mobile upload flows

Storage

Files are stored in **MinIO** (S3-compatible object storage). Metadata lives in the database.

Media Organization

- Media is attached to either a **job** or an **inspection response**
- Filenames are normalized to `{job_number}_{timestamp}.{ext}`
- Media is **soft-deleted** — can be recovered if accidentally removed

Customer Portal Access

Customer-facing media (photos, inspection findings) is served through an API proxy — customers never get direct access to internal storage URLs.

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17. Deferred Work

What Is Deferred Work?

When a customer declines or defers an estimate line, the system creates a **Deferred Work** record. This is work that was recommended but not approved.

For management, this is captured revenue. Deferred work represents jobs the customer didn't approve today — but might approve next visit, next month, or when reminded. Without tracking it, this revenue simply disappears.

What You Can Do with Deferred Work

Action	What Happens
Remind	Send the customer a reminder about the deferred item
Book into New Job	Create a new job from the deferred work
Close / Expire	Mark the deferred work as no longer relevant

Why It Matters for Revenue

Deferred work is **revenue you haven't captured yet**. The system keeps it visible so the team can follow up when the timing is right — next service visit, seasonal promotion, etc. Management can track the total deferred value and measure follow-up conversion over time.

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18. Notifications

How Notifications Work

1. An event occurs (inspection submitted, approval received, etc.)
2. A notification record is created in the database
3. If a webhook is configured, the notification is queued for delivery
4. Delivery is retried with exponential backoff on failure
5. If no webhook exists, the notification is marked as "sent" (no-op)

Database Is the Source of Truth

All notifications live in the database first. The delivery queue (BullMQ + Redis) is best-effort infrastructure. If Redis goes down, notifications still exist and can be re-queued when the system recovers.

Supported Events

- Inspection submitted
- Customer decision received

- Job status changed
- Approval link generated
- Deferred work reminder

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19. Settings & Configuration

Priority Matrix Settings

Customize the weight of every priority factor to match your workshop's priorities.

Access: Settings → Priority Matrix

Each factor has a weight slider (0–30). Changes take effect immediately on the job board.

Settings Groups:

Group	Factors
Promise Risk	Promise Overdue, Promise Due ≤ 2h, Promise Due ≤ 6h, No Promised Date
Customer	Customer Waiting, Customer Angry, Customer VIP, Customer Comeback
Approval	Waiting Customer Decision
Parts Risk	Parts Backorder, Parts Waiting Warehouse, Parts Need Order
Idle Risk	Idle 12h+, Idle 6h+
Stage Urgency	Stage: Checking/Diagnosis, Stage: QC/Near Delivery
Delivery	Ready to Inform
Value	High Estimate Value, Medium Estimate Value

Labour Rates

Set hourly rates for different operation types. These are used in the estimate builder to calculate labour line totals.

Users

Add, edit, deactivate, and assign roles to users.

Integrations

Configure webhook URLs for notification delivery (WhatsApp, SMS, email, etc.).

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20. Technical Architecture

System Components

```
┌──────────────┐      ┌──────────────┐      ┌──────────────┐
│   Browser    │─────▶│   Next.js    │─────▶│   NestJS     │
│   (User)     │      │   Web App    │      │   API        │
└──────────────┘      └──────┬───────┘      └──────┬───────┘
                             │                     │
                             │                     ├──────▶ MariaDB (Database)
                             │                     ├──────▶ Redis (Queue + Cache
                             │                     ├──────▶ MinIO (Media Storage
                             │                     └──────▶ BullMQ (Background J
                             │
                      ┌──────▼───────┐
                      │   Traefik    │
                      │   (Reverse   │
                      │    Proxy +   │
                      │    TLS)      │
                      └──────────────┘
```

Tech Stack

Component	Technology	Purpose
API	NestJS (Node.js)	Business logic, REST API
Web App	Next.js 16 (React)	Frontend UI
Database	MariaDB (MySQL)	All data storage
ORM	Prisma	Database access layer
Cache	Redis	Session cache, BullMQ backend
Queue	BullMQ	Background job processing
Media	MinIO (S3)	File and photo storage
Proxy	Traefik	Reverse proxy, TLS, routing
Auth	JWT + Refresh Tokens	User authentication

API Structure

- Base URL: `/api`
- Authentication: JWT Bearer tokens
- Health check: `/health` (no auth required)
- Customer portal: `/api/portal/*` (token-based auth, no JWT)
- Swagger docs: Available in non-production environments

Database Schema (Key Tables)

Table	Purpose
`users`	Staff accounts
`roles`	Permission roles
`customers`	Customer records
`vehicles`	Vehicle records
`jobs`	Main workshop job records
`job_status_history`	Audit trail of status changes
`inspections`	One inspection per job
`inspection_items`	Template items
`inspection_responses`	Actual findings
`estimate_lines`	Quote/estimate items
`estimate_line_history`	Audit trail of estimate changes
`approval_tokens`	Customer approval links (hashed)
`authorisation_decisions`	Per-line customer decisions
`deferred_work`	Declined/deferred estimate items
`media_files`	Uploaded media metadata
`notifications`	Queued and sent notifications
`settings`	System configuration
`labour_rates`	Hourly rates
`audit_logs`	Operational audit trail
`refresh_tokens`	Login sessions

21. Deployment Guide

Requirements

Requirement	Minimum
OS	Linux (Ubuntu 20.04+)
CPU	2 cores
RAM	4 GB
Disk	40 GB SSD
Docker	20.10+ with Compose V2
Domain	A registered domain with DNS access

Deployment Steps

1. **Clone the repository**

```
git clone https://github.com/hmelshenawy/superflow-api.git
cd superflow-api
git checkout dev
```

2. **Configure environment**

```
cp .env.example .env.production
# Edit .env.production with your settings
```

3. **Build and start**

```
docker compose --env-file .env.production up -d
```

4. **Set up TLS**
   - Traefik automatically provisions Let's Encrypt certificates
   - Point your domain DNS to the server IP
   - Traefik handles HTTP → HTTPS redirect
5. **Create admin user**
   - First user through the setup flow becomes admin

- Or seed via database

Environment Variables (Key)

Variable	Purpose
`DATABASE_URL`	MariaDB connection string
`JWT_SECRET`	Token signing key
`MINIO_ENDPOINT`	Media storage endpoint
`MINIO_ACCESS_KEY` / `MINIO_SECRET_KEY`	MinIO credentials
`CORS_ORIGINS`	Allowed frontend origins
`APP_DOMAIN`	Your domain (e.g., `prioraflow.com`)

Updating

```
git pull origin dev
docker compose --env-file .env.production build
docker compose --env-file .env.production up -d
```

Backups

- **Database:** Use `mysqldump` or MariaDB backup tools
- **Media:** Sync MinIO data directory
- **Config:** `.env.production` file should be securely stored

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22. Security Model

Authentication

- **Staff login:** JWT access tokens (short-lived) + Refresh tokens (long-lived)
- **Refresh tokens:** Stored as SHA-256 hashes, rotated on each use
- **Rate limiting:** Login (5/min), Refresh (10/min)
- **Session management:** View active sessions, revoke individually or all

Customer Portal

- **Token-based access:** Unique hashed token per approval link
- **No JWT required:** Customer accesses via secure link only
- **Token expiry:** 7 days default
- **Single use:** Token invalidated after decision submission
- **Media proxy:** Customer never accesses internal storage directly

Data Protection

- **Soft deletes:** Media and some records use soft deletion (recoverable)
- **Audit logging:** All significant actions are logged
- **Input validation:** Strict DTO validation rejects unknown fields
- **CORS:** Configured origins only
- **Helmet:** Security headers enabled
- **Throttling:** Global rate limiting on all API routes

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23. Glossary

Term	Definition
Job / RO	A repair order — the main unit of work in PrioraFlow
Priority Score	0–100 number indicating how urgently a job needs attention
Priority Level	Low / Normal / High / Critical — derived from the score
Workshop Stage	The physical stage of the vehicle inside the workshop
Parts Status	The supply chain state for parts needed by the job
Customer Informed	Flag indicating the customer has been told their vehicle is ready
Next Best Action	The system's recommendation for what the advisor should do next
Deferred Work	Estimate items the customer didn't approve — tracked for follow-up
Approval Token	A secure, hashed link sent to the customer for estimate approval
Inspection	A structured vehicle check performed by the technician
Estimate	The quote/price breakdown sent to the customer
DMS	Dealer Management System — the workshop's core business system

Term	Definition
Promised Date	The date/time the customer was told their vehicle would be ready
Idle Time	How long since the job was last updated
Kanban Board	Visual job board organized by status columns
Traefik	The reverse proxy that handles routing and TLS
MinIO	S3-compatible object storage for media files
BullMQ	Background job queue powered by Redis

PrioraFlow — Know what's urgent. Know what to do next.